TEAM ASSIGNMENT #3

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name:______________________________________

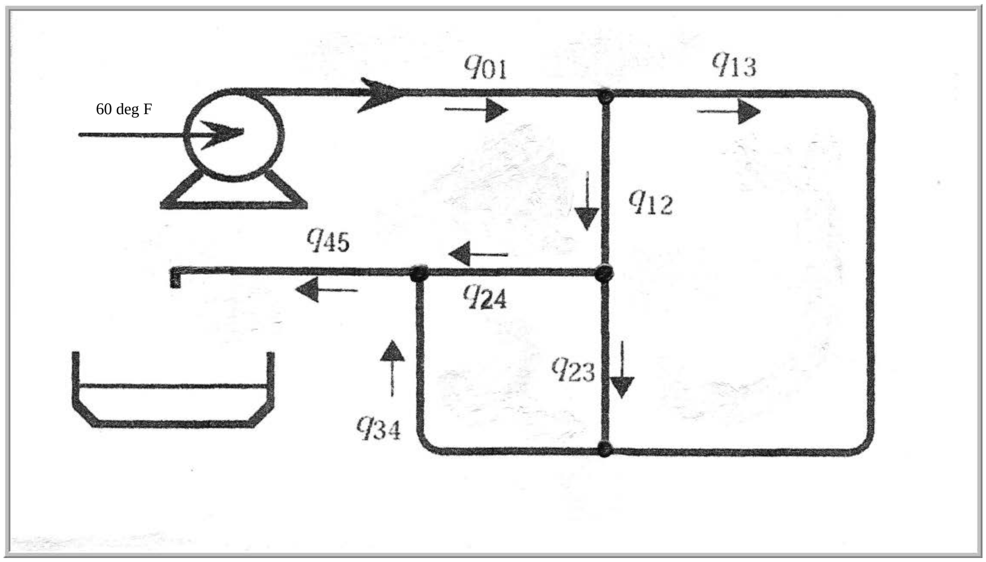
Due Date: Friday October 4th by 3:00pm in my office

Number of points..........................100
Your score...............................___

Note: **All problems are worth 100 points. You final score will be the average value**

Problem A: **Waste Water Treatment:**

Waste water is flowing with a total volumetric flow rate of $Q = 1500\,gpm$ in the pipeline network that is shown in the schematic below:



The waste water is at $60\,^{o}\mathrm{F}$ temperature and atmospheric pressure and it is discharged at the end of the pipeline network in an open pool that is subsequently used for reclamation. All the pipe segments in the network are 6-inch schedule 40 carbon steel. The equivalent lengths of the pipes connecting the different nodes are as follows:

$$L_{01} = 1500\,m$$
$$L_{12} = L_{23} = L_{45} = 300\,m$$
$$L_{13} = 2000\,m \quad L_{24} = 1200\,m \quad L_{34} = 1500\,m$$

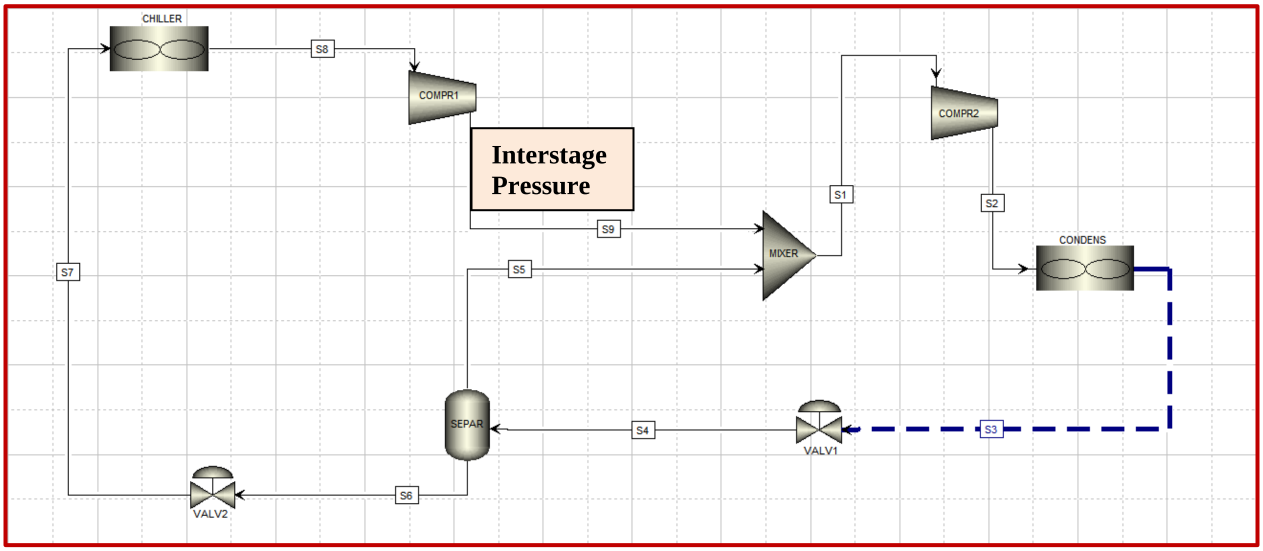
Create an ASPEN Plus simulation to calculate the minimum horsepower requirement of the pump, assuming 75% efficiency. There is no starting ASPEN Plus file for this problem. You have to create your own.

Problem B: **Optimal Diameter of a Steam Pipe.**

$774\,SCFM$ of low pressure saturated steam is to be transported in a steam pipeline with an average lifetime roughness of $\varepsilon = 0.5\,mm$. The boiler will be operating at a pressure range $\left[1\,bar \text{ to } 20\,bar\right]$. Develop a model for the calculation of the optimal steam pipe diameter as a function of the boiler pressure. Capital and operating cost data are given in the article **SteamPipe.pdf** which has been uploaded on the Enhanced Learning section of the UIC Blackboard along with a starting ASPEN Plus template: **SteamPipeTemplate.bkp**

Problem C: Refrigeration with Flash Tank Economizer and Two Stages of Compression

We wish to remove $3MW$ from a process gas stream in a chiller operating at $-35\,^{o}C$ and reject it to the environment in a condenser operating at $35\,^{o}C$ using propane as the working fluid. Conduct a sensitivity analysis and plot the total power requirement as a function of the interstage pressure. Using the Optimization Flowsheeting option determine the optimal interstage pressure for the cycle. What is the COP of the cycle for the optimal pressure and how does it compare with the maximum thermodynamic COP. Assume that both compressors operate at 75% isentropic efficiency. Starting ASPEN Plus template: **RefrigerationTemplate.bkp**



Problem D: Cyclohexane production

For the cyclohexane production process of **Assignment #2**, prepare design specifications to meet the target ratios for hydrogen and liquid recycle and devise a methodology using the ASPEN Plus Flowsheeting options to calculate the raw material flow rates for a production target of 100 KMTA. Prepare a **Custom Table** displayed on the flowsheet that would show the results of your methodology. Starting ASPEN Plus template: **CHProductionTemplate.bkp**